

Introduction to Python

Author: Luv, Juan

Description:

Introduction to Python is the perfect starting point for anyone eager to learn Python programming. This beginner-friendly book introduces Python's syntax and basic concepts in a clear and approachable manner. Covering essential topics, it helps readers build a strong foundation for programming and problem-solving.

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1. INTRODUCTION TO PYTHON

Python is a high-level, interpreted programming language known for its simplicity and readability. It is widely used in web development, data science, artificial intelligence, automation, and more.

Python focuses on clean and readable code, making it ideal for beginners.

2. SETTING UP PYTHON ENVIRONMENT

To start programming in Python, you need:

- Python installed (python.org)
- A code editor (VS Code recommended)

Check installation:

```
python --version
```

Your first program:

```
print("Hello, Python!")
```

3. VARIABLES AND DATA TYPES

Variables store data values.

Example:

```
name = "John"
age = 25
height = 1.75
is_student = True
```

Data types:

- String
- Integer
- Float
- Boolean

4. INPUT AND OUTPUT

Python allows interaction with users.

Example:

```
name = input("Enter your name: ")
print("Hello", name)
```

5. OPERATORS IN PYTHON

Operators perform operations on variables.

Example:

```
a = 10
b = 5
print(a + b)
print(a > b)
```

Types:

- Arithmetic
- Comparison
- Logical

6. CONTROL FLOW

Used for decision-making.

Example:

```
age = 18
if age >= 18:
    print("Adult")
else:
    print("Minor")
```

7. LOOPS

Loops repeat actions.

For loop:

```
for i in range(5):
    print(i)
```

While loop:

```
i = 0
while i < 5:
```

```
print(i)
i += 1
```

8. FUNCTIONS

Functions are reusable blocks of code.

Example:

```
def greet(name):
    return "Hello " + name
print(greet("Juan"))
```

9. LISTS AND TUPLES

List:

```
fruits = ["Apple", "Banana", "Mango"]
```

Tuple:

```
colors = ("Red", "Green", "Blue")
```

Lists are mutable, tuples are not.

10. DICTIONARIES AND SETS

Dictionary:

```
user = {
    "name": "John",
    "age": 25
}
```

Set:

```
numbers = {1, 2, 3, 4}
```

11. FILE HANDLING

Writing to a file:

```
file = open("data.txt", "w")
file.write("Hello Python")
file.close()
```

Reading a file:

```
file = open("data.txt", "r")
print(file.read())
file.close()
```

12. ERROR HANDLING

Used to handle errors safely.

Example:

try:

```
print(10 / 0)
```

except ZeroDivisionError:

```
print("Cannot divide by zero")
```

13. INTRODUCTION TO MODULES

Modules are pre-written Python files.

Example:

```
import math  
print(math.sqrt(16))
```

14. MINI PROJECT

Simple calculator:

```
def add(a, b):  
    return a + b  
def subtract(a, b):  
    return a - b  
print(add(5, 3))  
print(subtract(5, 3))
```

15. CONCLUSION

Python is one of the easiest and most powerful programming languages to learn.

By mastering its basics, you open doors to fields like data science, AI, web development, and automation.

The key to success is practice and building real projects consistently.